

Schrödinger Bridge for Non-linear ROMs

Presentation of Progress

June 10, 2026

1D Resolution: Heat Kernel & Method of Images

The Heat Semigroup

The Schrödinger Bridge (IPFP) relies on solving the heat equation $\partial_t \varphi = \gamma \Delta \varphi$. The analytic solution is given by the heat kernel integral:

$$P_t f(x) = \int_{\Omega} K_t(x, y) f(y) dy \quad \text{where} \quad K_t(x, y) = \frac{1}{\sqrt{4\pi\gamma t}} \exp\left(-\frac{(x-y)^2}{4\gamma t}\right)$$

Method of Images (Neumann Boundary Conditions)

To prevent mass from leaking outside our finite 1D domain $[-L, L]$, we enforce zero-flux boundary conditions using the Method of Images. We sum the direct kernel and reflected "ghost" kernels:

$$K_t^{Neumann}(x, y) = \sum_{k=-N}^N [K_t(x, y + 2kL) + K_t(x, -y + 2kL - 2L)]$$

1D Transport Cases: Gaussian to Gaussian

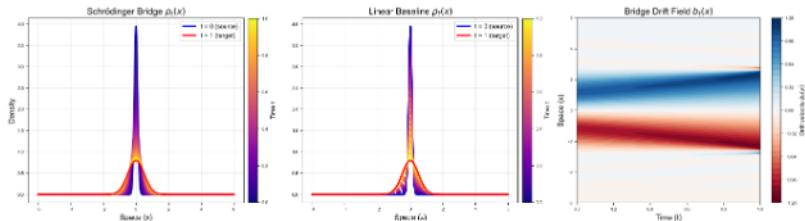


Figure: Density Evolution Dashboard

1D Transport Cases: Gaussian to Gaussian

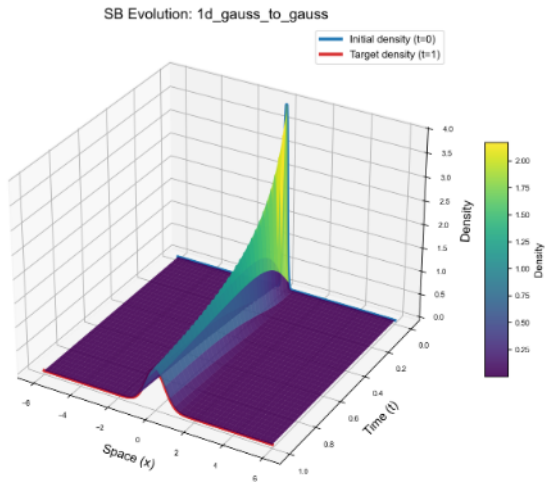


Figure: Density Evolution

1D Transport Cases: Square to Gaussian

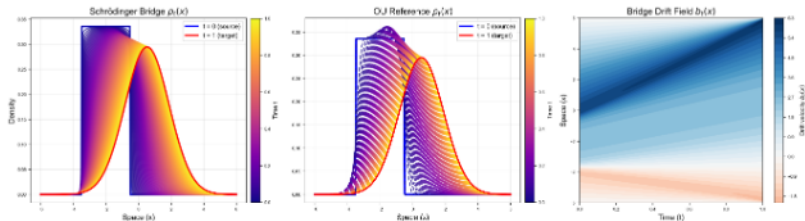


Figure: Density Evolution Dashboard

1D Transport Cases: Square to Gaussian

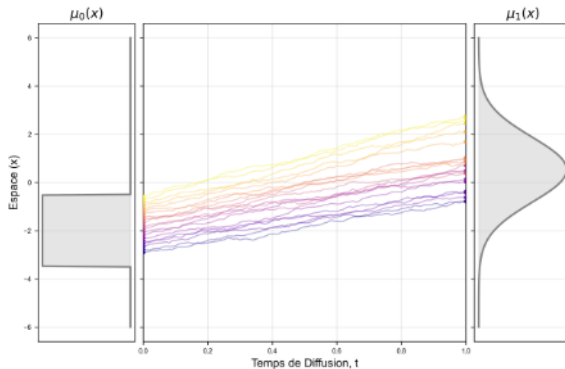


Figure: SDE Particle Trajectories

1D Transport Cases: Bimodal Target

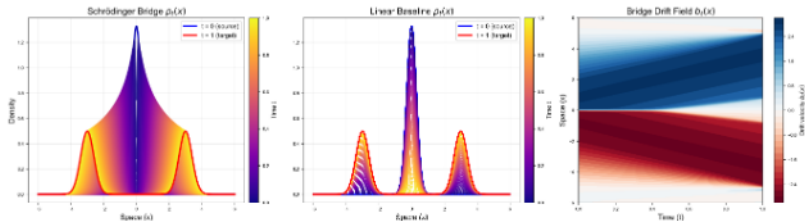


Figure: Density Evolution Dashboard

1D Transport Cases: Bimodal Target

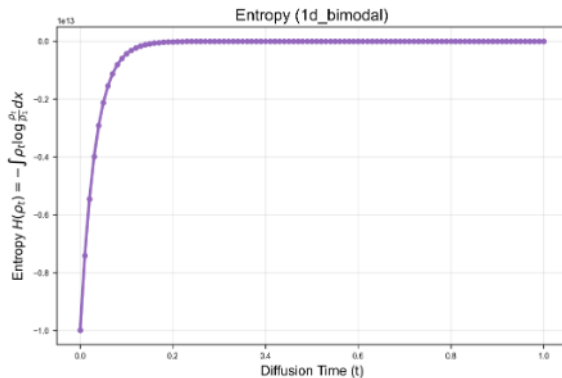


Figure: Entropy Evolution

1D Transport Cases: Square to OU

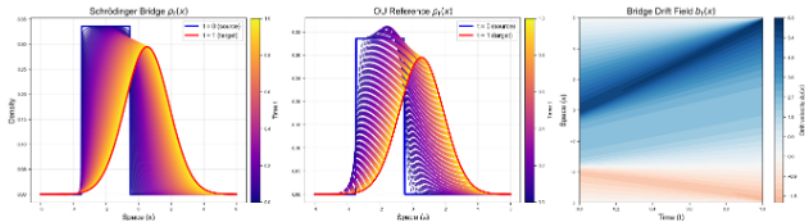


Figure: Density Evolution Dashboard

1D Transport Cases: Square to OU

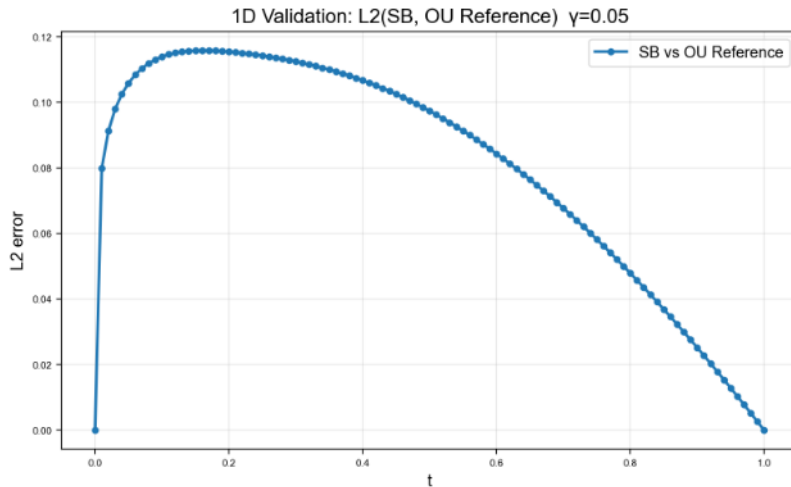


Figure: SB vs OU error

2D Resolution: Unstructured FVM Heat Solver

FVM Semigroup on Unstructured Meshes

To solve $\partial_t \varphi = \gamma \Delta \varphi$ on 2D aerodynamic meshes (e.g., bump, diamond body):

- **Discretization:** Two-Point Flux Approximation (TPFA) using cell barycenters.
- **Time Integration:** Explicit SSP-RK2 (Heun's method).

Anderson Acceleration

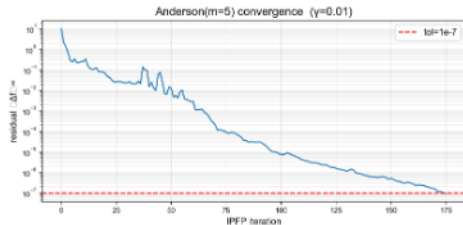


Figure: Anderson Acceleration Convergence

2D Transport Cases: Gaussian to Gaussian

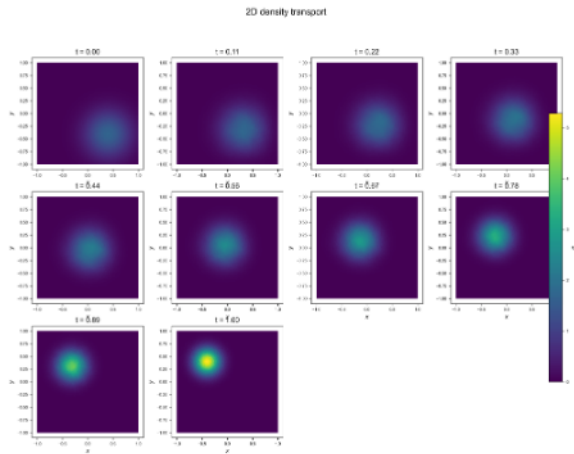


Figure: Unstructured Density Transport

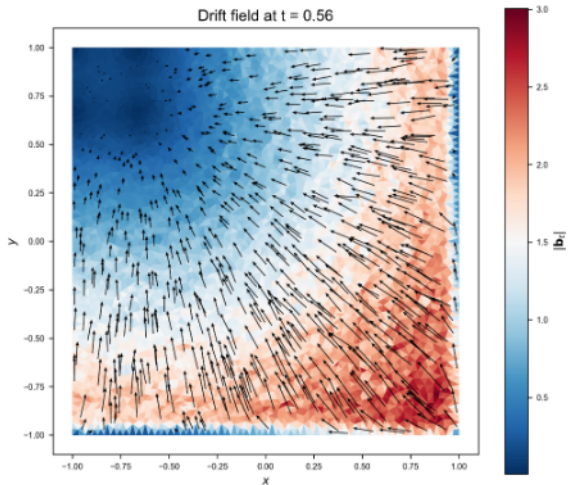


Figure: Drift Vector Field $b_t(x)$

2D Transport Cases: Bimodal Splitting

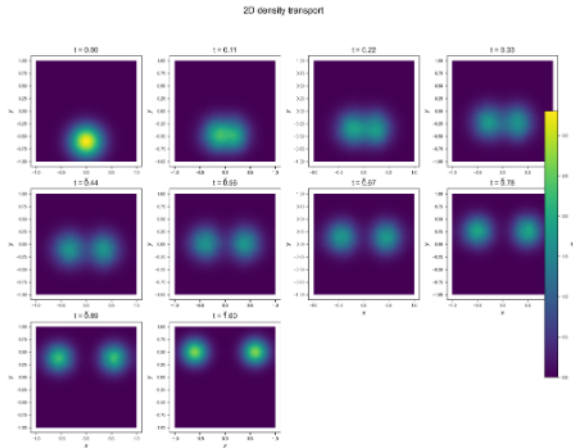


Figure: Unstructured Density Transport

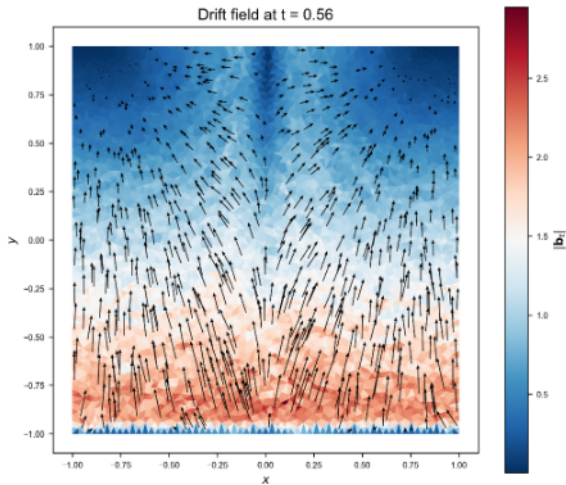
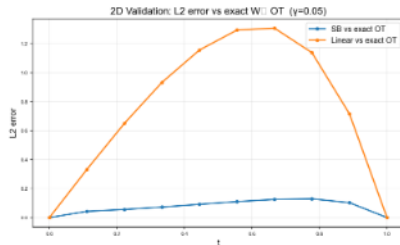
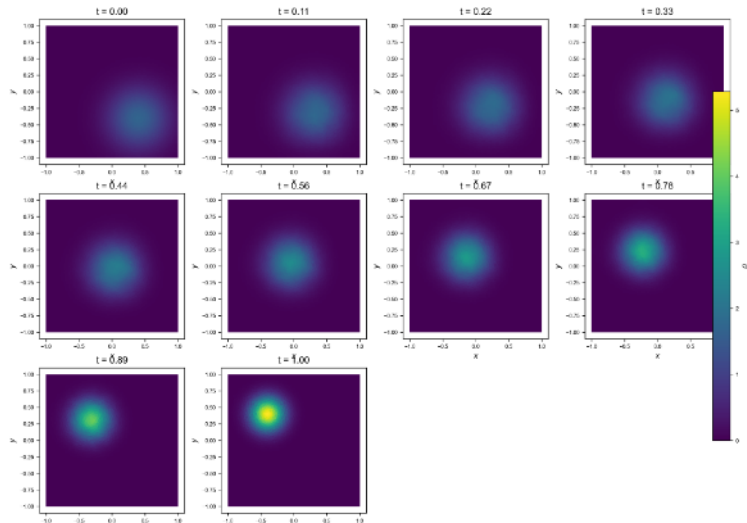


Figure: Drift Vector Field $b_t(x)$

Validation: Gaussian $\rightarrow$ Gaussian ($\gamma = 0.05$)

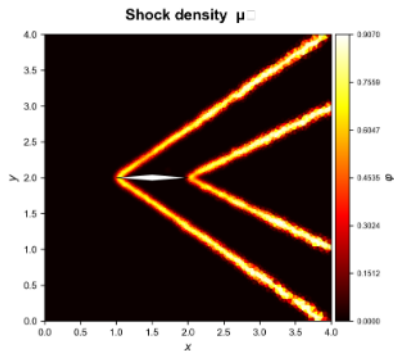
2D density transport



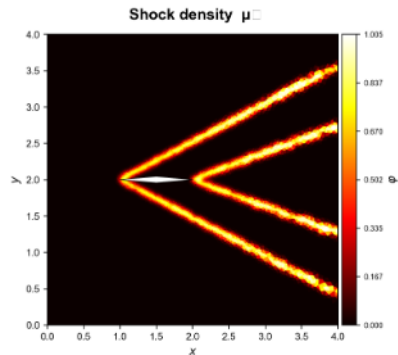
L_2 error vs. exact W_2 geodesic

$$\rho_t^{OT} = \mathcal{N}(c_t, \sigma_t^2 I)$$

Mach Application: Shock Probability Densities



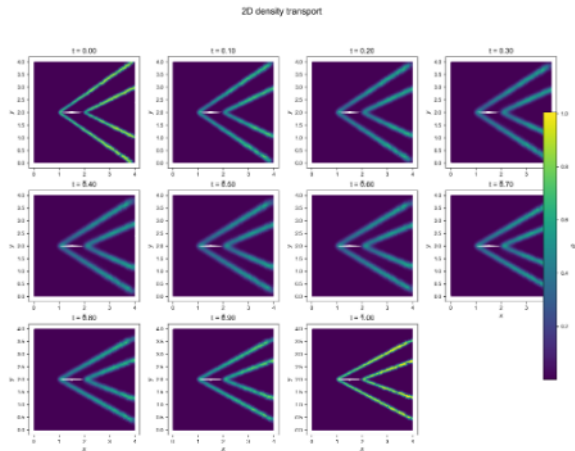
μ_0 — Mach 2.00



μ_1 — Mach 2.50

Ducros sensor $\rightarrow$ binary shock mask $\rightarrow$ FVM heat smoothing $\rightarrow$ normalise.

SB Density Transport: $\mu_0 \rightarrow \mu_1$



CDI: From IPFP Potentials to Mach Interpolation

1 IPFP $\rightarrow$ Sinkhorn potentials f, g

$$\mu_0 = e^f \cdot P_1 e^g, \quad \mu_1 = e^g \cdot P_1 e^f, \quad P_t \varphi = \text{FVM heat semigroup}$$

2 Barycentric transport map (3 heat solves on e^g)

$$T(x_0) = \frac{P_1[\text{id} \cdot e^g](x_0)}{P_1[e^g](x_0)}, \quad X(t, x_0) = (1 - t)x_0 + t T(x_0)$$

3 Two-map CDI reconstruction

Backward map Φ_0 (fixed-point, 6 steps):

$$\Phi_0(x) = x_0^* : X(t, x_0^*) = x$$

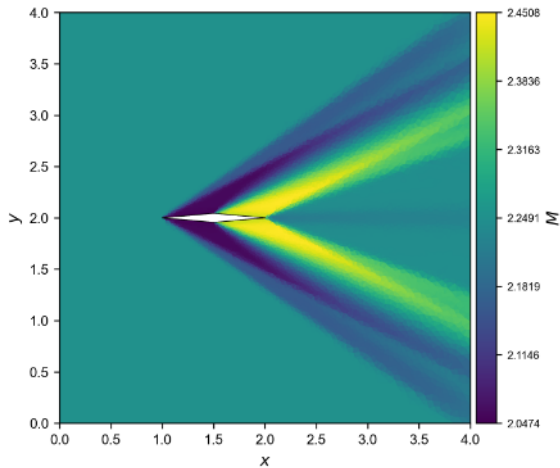
Forward map Φ_1 (symmetry $f \leftrightarrow g$, then T):

$$\Phi_1(x) = T(x_0^*)$$

$$\hat{M}_t(x) = (1 - t) M_0(\Phi_0(x)) + t M_1(\Phi_1(x))$$

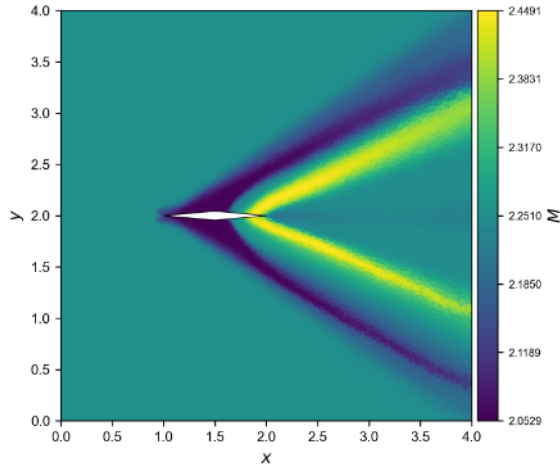
Mach Interpolation at $t = 0.5$

Linear $t=0.50$



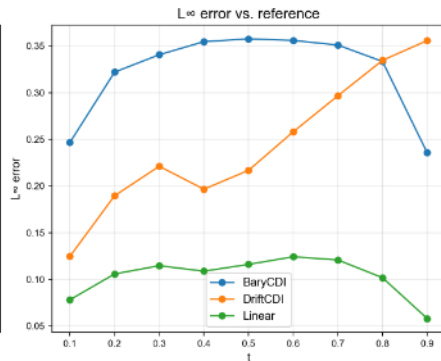
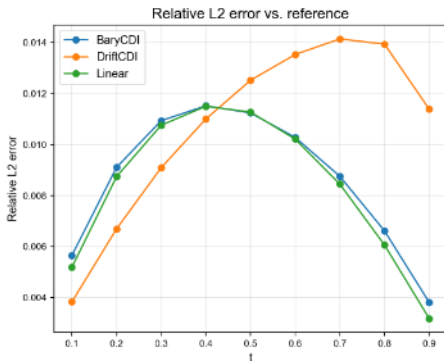
Linear

SB-CDI (barycentric) $t=0.50$



Barycentric CDI

Error vs. High-Resolution Reference



Relative L_2 and L_∞ errors at $t = 0.1, \dots, 0.9$ against solver solutions at intermediate Mach numbers.